

29 For $u, v \in V$, define $d(u, v) = \|u - v\|$

- (a) Show that d is a metric on V .
- (b) Show that if V is finite-dimensional, then d is a complete metric on V (meaning that every Cauchy sequence converges).
- (c) Show that every finite-dimensional subspace of V is a closed subset of V .

- (a) In order to show that d is a metric, we need to show that it is positive definite, symmetric, and satisfies the triangle inequality.

First we show that d is positive definite. Let u, v, w be any vectors in V . Since $\|w\| = 0$ if and only if $w = 0$, we have that $d(u, v) = 0$ if and only if $u - v = 0$. Thus, we have $d(u, v) = 0$ if and only if $u = v$. Note also that $\|w\| \geq 0$, giving us $d(u, v) \geq 0$ always. Combining these two statements about d gives us that it is positive definite.

Now we show that d is symmetric. This follows obviously from the properties of the norm of a scaled vector. For any $u, v \in V$

$$d(u, v) = \|u - v\| = \|(-1)(v - u)\| = |-1|\|v - u\| = \|v - u\| = d(v, u).$$

Finally, we only need to point out an obvious conversion between what the triangle means for norms and what it means for metric spaces. For any $u, v, w \in V$

$$d(u, v) + d(v, w) = \|v - u\| + \|w - v\|.$$

By the triangle inequality (for norms) we have

$$\|v - u\| + \|w - v\| \leq \|(v - u) + (w - v)\| = \|w - u\|.$$

Thus, we have the triangle inequality for metrics:

$$d(u, v) + d(v, w) \leq d(u, w)$$

- (b) If V is finite-dimensional, then we have an orthonormal basis $u_1, \dots, u_m$. (This is proven later in the book). Let $\{v_n\}$ be any Cauchy sequence in V .

Then we can write each element of the sequence

$$v_n = a_{n,1}u_1 + \dots + a_{n,m}u_m.$$

Since $\{v_n\}$ is Cauchy, we have that for any $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that for all $j, k > N$ we have $d(v_j, v_k) < \varepsilon$. We can rewrite the expression as

$$d(a_{j,1}u_1 + \dots + a_{j,m}u_m, a_{k,1}u_1 + \dots + a_{k,m}u_m) = \|(a_{j,1} - a_{k,1})u_1 + \dots + (a_{j,m} - a_{k,m})u_m\|.$$

Thus,

$$\|(a_{j,1} - a_{k,1})u_1 + \dots + (a_{j,m} - a_{k,m})u_m\| < \varepsilon$$

(with all the same qualifiers for j, k, ε). Since the basis vectors are orthogonal, and the right and left hand sides of the inequality are obviously nonnegative, we get

$$|a_{j,1} - a_{k,1}|^2 \|u_1\|^2 + \dots + |a_{j,m} - a_{k,m}|^2 \|u_m\|^2 < \varepsilon^2$$

and since the basis vectors are also unit vectors, we can simplify this to

$$|a_{j,1} - a_{k,1}|^2 + \dots + |a_{j,m} - a_{k,m}|^2 < \varepsilon^2$$

Since all the terms on the left hand side are nonnegative, we can write

$$|a_{j,p} - a_{k,p}|^2 < \varepsilon^2$$

for any $p \in \{1, \dots, m\}$. Again, since both sides of the inequality are positive, we get

$$|a_{j,p} - a_{k,p}| < \varepsilon.$$

Specifically, for any $p \in \{1, \dots, m\}$, for any $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that for all $j, k > N$

$$|a_{j,p} - a_{k,p}| < \varepsilon.$$

That is, for any $p \in \{1, \dots, m\}$ the sequence $\{a_{n,p}\}$ is Cauchy.

Since V is a field over $\mathbb{R}$ or $\mathbb{C}$, both of which are complete, for all p , $\{a_{n,p}\}$ converges to some A_p . That is, for any $\varepsilon' > 0$, there is some $N \in \mathbb{N}$ such that for all $n > N$, $|A_p - a_{n,p}| < \varepsilon'$. This motivates us to think about the sequence of vectors converging to $A_1u_1 + \dots + A_mu_m$.

For any $\varepsilon' > 0$ we can have some N such that for all $n > N$, for all p , $|A_p - a_{n,p}| < \frac{\varepsilon'}{m}$ (if for each sequence of coefficients we get N_p , then take some n greater than $\max N_p$). We consider

$$d(A_1u_1 + \dots + A_mu_m, a_{n,1}u_1 + \dots + a_{n,m}u_m) = \|(A_1u_1 + \dots + A_mu_m) - (a_{n,1}u_1 + \dots + a_{n,m}u_m)\|$$

and perform the same expansion as previously to get

$$d(A_1u_1 + \dots + A_mu_m, a_{n,1}u_1 + \dots a_{n,m}u_m) \leq |A_1 - a_{n,1}| + \dots + |A_m - a_{n,m}| < m \frac{\varepsilon'}{m} = \varepsilon'$$

Specifically, for any $\varepsilon' > 0$ there is some $N \in \mathbb{N}$ such that for all $n > N$

$$d(A_1u_1 + \dots + A_mu_m, a_{n,1}u_1 + \dots a_{n,m}u_m) < \varepsilon'.$$

That is, $\{v_n\}$ converges to $A_1u_1 + \dots A_mu_m$.

- (c) By the construction in part (b), any Cauchy sequence in a subspace U of V converges to a point in U . Thus, U contains all of its limits points, and thus, is closed. ■

30 Fix a positive integer n . The **Laplacian** Δp of a twice differentiable function p on $\mathbb{R}^n$ is the function on $\mathbb{R}^n$ defined by

$$\Delta p = \frac{\partial^2 p}{\partial x_1^2} + \dots + \frac{\partial^2 p}{\partial x_n^2}$$

A function p is called **harmonic** if $\Delta p = 0$.

A **polynomial** on $\mathbb{R}^n$ is a linear combination of functions of the form $x_1^{m_1} \dots x_n^{m_n}$, where $m_1 \dots m_n$ are nonnegative integers.

Suppose q is a polynomial on $\mathbb{R}^n$. Prove that there exists a harmonic polynomial p on $\mathbb{R}^n$ such that $p(x) = q(x)$ for every $x \in \mathbb{R}^n$ with $\|x\| = 1$.

Let $\mathcal{P}(\mathbb{R}^n)$ denote the space of polynomials on $\mathbb{R}^n$ and let $\mathcal{P}_m(\mathbb{R}^n)$ be the space of polynomials on $\mathbb{R}^n$ of degree at most m . If $q \in \mathcal{P}(\mathbb{R}^n)$, and we want some harmonic polynomial p such that $p = q$ for every $x \in \mathbb{R}^n$ with $\|x\| = 1$, we can reasonably think that

$$p = q + (1 - \|x\|^2)r$$

where r is some polynomial. Such a p would necessarily give us $p = q$ at all the desired points, but there might not be such a harmonic polynomial. Thus, we need to show that there is some harmonic polynomial of the same form. That is, we need to show that there exists some r such that

$$\Delta p = \Delta q + \Delta((1 - \|x\|^2)r) = 0.$$

This amounts to showing that there exists some r such that

$$\Delta((1 - \|x\|^2)r) = -\Delta q.$$

This, can be done by considering the linear map giving $r \mapsto \Delta((1 - \|x\|^2)r)$ and showing that it is surjective. However, since the space of all polynomials is infinite-dimensional,

this would require a direct proof and would not amount to any simplification. Instead, we can choose the right finite-dimensional space so that the map is an operator and we only have to prove that it is injective.

Note that if $\deg q = m + 2$, then $\deg \Delta q = m$. Similarly, if $\deg r = m$, then $\deg(1 - \|x\|^2)r = m + 2$ and thus $\deg \Delta(1 - \|x\|^2)r = m$. Thus, we can just consider the space of polynomials of degree at most m .

Specifically, we can consider $T \in \mathcal{L}(\mathcal{P}_m(\mathbb{R}^n))$ defined by

$$Tr = \Delta(1 - \|x\|^2)r.$$

To be done, we only have to show that T is injective.

Suppose $Ts = 0$. Then, using the product rule for the Laplacian we have

$$s\Delta(1 - \|x\|^2) + (1 - \|x\|^2)\Delta s + \nabla s \cdot \nabla(1 - \|x\|^2) = 0.$$

Calculating all the derivatives where we can, we get

$$s(-2n) + (1 - \|x\|^2)\Delta s + \sum_{j=1}^n s_{x_j}(-2x_j) = 0$$

which can be rewritten as

$$s = \frac{1}{2n}(1 - \|x\|^2)\Delta s - \frac{1}{n} \sum_{j=1}^n \frac{\partial s}{\partial x_j} x_j$$

■

31 Use inner products to prove Apollonius's identity: in a triangle with sides of length a , b , and c , let x be the length of the line segment from the midpoint of the side of length c to the opposite vertex. Then

$$a^2 + b^2 = \frac{1}{2}c^2 + d^2$$

Consider the obvious choice of v_a , v_b , v_c , and v_d such that $\|v_\alpha\| = \alpha$. Note that this then gives us

$$\frac{v_c}{2} + v_d = v_a = v_c + v_b$$

and

$$\frac{v_c}{2} + v_b = v_d.$$

Thus, we have

$$\left\| \frac{v_c}{2} + v_d \right\|^2 = \|v_a\|^2$$

which we can expand using the dot product to get

$$\frac{c^2}{4} + d^2 + \frac{v_c}{2} \cdot v_d = a^2.$$

We also have

$$\left\| \frac{v_c}{2} + v_b \right\|^2 = \|v_d\|^2$$

which gives us

$$\frac{c^2}{4} + b^2 + \frac{v_c}{2} \cdot v_b = d^2.$$

Swapping sides and adding the equations gives us

$$2d^2 + \frac{v_c}{2} \cdot v_d = a^2 + b^2 + \frac{v_c}{2} \cdot v_b$$

which we can rewrite as

$$2d^2 + \frac{v_c}{2} \cdot (v_d - v_b) = a^2 + b^2$$

$$2d^2 + \frac{v_c}{2} \cdot v_c = a^2 + b^2$$

which finally gives us

$$a^2 + b^2 = 2d^2 + \frac{c^2}{2}$$

as desired. ■